

CO2 AT3 – Class Test

DSA 0201 – Computer Vision with Open cv

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Question 1: Image Preprocessing and Feature Extraction

a) Preprocessing Pipeline

To improve the quality of farmland images before pest detection, the following preprocessing pipeline can be used.

Step 1: Noise Reduction

Method: Gaussian Filter

- Removes Gaussian noise caused by the drone sensor and wind movement.
- Produces a smoother image while preserving important crop details.

Step 2: Illumination Correction

Method: Histogram Equalization

- Corrects uneven lighting caused by changing sunlight and shadows.
- Makes the brightness more uniform across the image.

Step 3: Contrast Enhancement

Method: Contrast Limited Adaptive Histogram Equalization (CLAHE)

- Enhances local contrast.
- Makes pest-affected regions more visible.
- Improves the visibility of texture and colour differences.

Step 4: Image Normalization

Method: Min-Max Normalization

- Converts pixel values to a common intensity range (0–1 or 0–255).

- Ensures consistent image quality.
- Improves the performance of feature extraction algorithms.

b) Feature Extraction

Selected Method: Local Binary Pattern (LBP)

Why LBP is Suitable

- LBP is an effective texture feature extraction method.
- Pest-damaged crops show different textures compared to healthy crops.
- LBP captures small texture variations accurately.
- It is simple, fast, and works well for aerial farmland images.
- It helps distinguish healthy and pest-affected crop regions.

c) Rotation and Scale Variations

Effect of Rotation and Scale

Drone images are captured from different heights and angles.

This can cause:

- Features to appear rotated.
- Crop regions to appear larger or smaller.
- The same pest-affected area may look different in different images.
- Feature matching becomes less accurate.

Methods to Improve Robustness

1. SIFT (Scale-Invariant Feature Transform)

- Detects features that remain stable even if the image is rotated or resized.
- Provides reliable feature matching under different viewing conditions.

2. ORB (Oriented FAST and Rotated BRIEF)

- Rotation invariant.
- Faster than SIFT.
- Suitable for real-time drone image analysis.

Conclusion

Using SIFT or ORB makes feature extraction more robust because these methods can detect the same crop features even when the drone captures images from different heights or viewing angles.

Question 2: Stereo Vision and Motion Estimation

a) Stereo Disparity and Depth Estimation

Stereo cameras consist of two cameras (left and right) placed a small distance apart. Each camera captures the same scene from a slightly different angle.

Working:

1. The left and right cameras capture images simultaneously.
2. The system identifies the same object in both images.
3. It measures the difference in the object's position between the two images, called disparity.
4. Using the disparity value, the system calculates the depth (distance) of the object.

Disparity for Different Objects

- **Nearby objects:** Large disparity because the object appears at very different positions in the left and right images.
- **Distant objects:** Small disparity because the object appears almost at the same position in both images.

Conclusion

- Large disparity → Object is closer.

- Small disparity → Object is farther away.

This helps the self-driving car estimate the distance of vehicles and pedestrians to avoid collisions.

b) Optical Flow for Motion Tracking

Optical flow estimates the movement of objects between two consecutive video frames.

Working:

1. Capture two consecutive video frames.
2. Detect feature points on the pedestrian.
3. Track the movement of these feature points between the frames.
4. Generate optical flow vectors showing the direction and speed of movement.
5. Continuously update the pedestrian's position.

If the Pedestrian Changes Speed or Direction

- The optical flow vectors change in length and direction.
- Longer vectors indicate faster movement.
- Shorter vectors indicate slower movement.
- This helps the self-driving car continuously track the pedestrian and predict movement.

c) Errors and Mitigation Techniques

1. Camera Misalignment

- Left and right images do not match correctly.
- Causes incorrect depth estimation.
- Perform camera calibration to align both cameras accurately.

2. Poor Lighting

- Objects become difficult to detect.

- Optical flow and stereo matching become less accurate.
- Apply Histogram Equalization or CLAHE to improve image brightness and contrast.

3. Occlusion

- One object blocks another.
- Some feature points disappear, causing tracking errors.
- Use a Kalman Filter to predict the object's position until it becomes visible again.

4. Fast Motion

- Large movement between frames causes incorrect optical flow estimation.
- Use the Pyramidal Lucas-Kanade Optical Flow method to handle large movements.

5. Texture less Regions

- Plain surfaces provide very few feature points.
- Depth estimation becomes unreliable.
- Detect strong feature points using methods such as SIFT or ORB for more reliable matching.